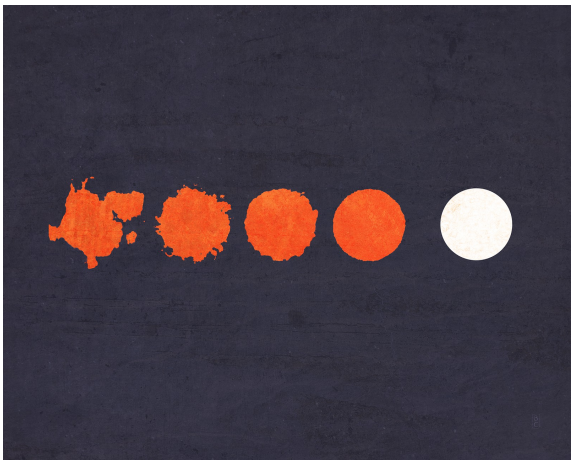
Brilliance and Potential:
Principles and Practices of Mastery Learning

Benjamin's Brilliance

"I have, on my office wall, a set of four paintings. The particular genius of these paintings is that most students don't notice them."
--Letter to self, ¶ 1, Feb. 2021



Benjamin had a problem.

Now, I've had some problems. Big ones, too.

I mean, I'm talking 1.21 gigawatt, flux-capacitor-powering, warp-core breaching, avatar-state-inducing, lock-S-foils-in-attack-position, time-to-drink-the-gummi-berry-juice level problems.

You've probably had some too.

Neither yours nor mine, however, were anything like Benjamin's problem.

My problems do not typically have their own wikipedia page, for one thing. The fabulously-first-named Benjamin (no relation) had a problem of the wikipedia-page-variety. You see, his problem was this: all of his students were brilliant.

Good problem, right?

Benjamin was an educational researcher. He believed that more students were brilliant than we tend to think, if given the right conditions.

He was right. From the afore-mentioned wikipedia page: The average student in his class "was above 98% of the students in the control class."

Suddenly the average student is in the 98th percentile. Think on that for a moment.

A similar statistic: 90% of students' performance levels were at or above that "reached by only the highest 20%" of the control section of the course. In other words, under his system, 90% of students were top-quintile students.

Bloom's system was simple: he replaced the traditional 25-to-1 student to teacher ratio with a 1-to-1 tutor program. Knowing that he couldn't replicate this at scale, he created a new system based on personalization.

He called it “mastery learning.”

I love mastery learning. I think it works. The research suggests it does, and my personal experience matches. One of my goals in this essay is to describe it, and hopefully help you apply it in small, bite-sized chunks.

It’s not my most important goal, however.

You will likely see some of these ideas and think, “Well, that’s just good teaching!” True enough. In fact, you may notice that what I am describing is simply a repackaging of the principles of Christ-like teaching.

Indeed.

That. That is the most important part:

To the extent that any innovation, reform, or principle works--including mastery learning--it does so because it conforms to eternal law. We succeed when we believe all students can succeed under the right conditions, and that we have the power to create those conditions. To the extent mastery learning helps students, it is because we are teaching as The Savior would, using the principles that He did.

Programs without principles are just colors on a page. To make art, the two must be inseparably connected. I have organized this essay into four parts, each devoted to a major principle of mastery learning. I’ve attempted to tie together practice and principle so that you can try them out for yourself.

Principle 1: Rigor. “I Excuse Not Myself.”
“Isaiah speaks of the difference between seeing and perceiving. Every student sees the paintings. Few perceive. Few grasp the principles behind the paint. The ones who do are willing to pay the required price: They seek. They ask. They knock.”
--Letter to self, ¶ 2, Feb. 2021

I asked students, on a whim, whether they preferred easier classes or more rigorous ones. It was an anonymous zoom poll, so I was pretty sure what the answer would be.

They surprised me. By a more than two-to-one ratio, they answered that they preferred rigor.

I regularly teach my students that, as a general rule, the higher a teacher’s expectations, the more their students will achieve. This is backed by a wealth of research: those teachers (and parents!) who are able to

balance being loving and warm, while at the same time holding high expectations for their children are precisely the ones that see the best outcomes for students.

High expectations communicate respect. They also communicate potential. When The Savior says “be ye therefore perfect” He’s telling you what He expects--and that implies that you can do it. (See: 1 Ne. 3:7.)

I cannot think of one scriptural instance of the Lord apologizing for his high standards.I cannot think of a higher-rigor task than living up to the divine potential within us.

Mastery learning is about rigor--and the kind of rigor that Doug Lemov would call“without apology” rigor. Mastery learning is all about setting challenging targets, and helping students reach them; more importantly, it reframes education away from compliance, and toward achievement. The reward for mastering hard material is the feeling of satisfaction from mastering hard material--and the joy that comes from learning.

That’s the principle. Now for some practices.

Practices:

Rigor Practice 1:
Binary Grading: The “Try Again”
I never liked giving C’s. I recently switched to binary grading (what I sometimes call “A-or-nothing” grading): if it’s mediocre, if it’s sub-par, student gets a zero--with the push to “try again, we’re rooting for you!” Allowing resubmission lets me set a higher bar than any one-shot assignment. (One student recently complained to me “I miss getting a C. This is way more work!”)

Rigor Practice 2:
More Time At-Bat
Doug Lemov, in *Teach Like a Champion*, explains that students learn the most from being at-bat; not by studying physics, or watching tapes of good batters swinging. Talking with a friend recently about how to develop writing skills, we talked about switching out a midterm and final 15-page essay for weekly 1-page essays. Same amount of grading, more chances to be at-bat, and a feedback-improvement cycle that is faster and more focused.

Art Note: The title of the first piece of art is “Come Unto Christ.” I can hardly see it without finishing the phrase in my mind, “and be perfected in Him.” It speaks to me of the difficult--and eternally worthy--task of becoming something more than we are.

Principle 2: Instructional Clarity. “I Glory in Plainness.”
“You may have noticed, as I have, that when students come into my office to talk about grades, assignments, and due dates, they are often there to talk about much weightier matters than grades, assignments, and due dates. They come to talk about grace, growth, love, meaning, and life.”
--Letter to self, ¶ 3, Feb. 2021

Let me begin on a note of clarity (which seems appropriate to this section of the essay): the same students who voted for more rigor stipulated that it come with less busy work. They told me that less is more, and the best faculty are the most instructionally-focused faculty.

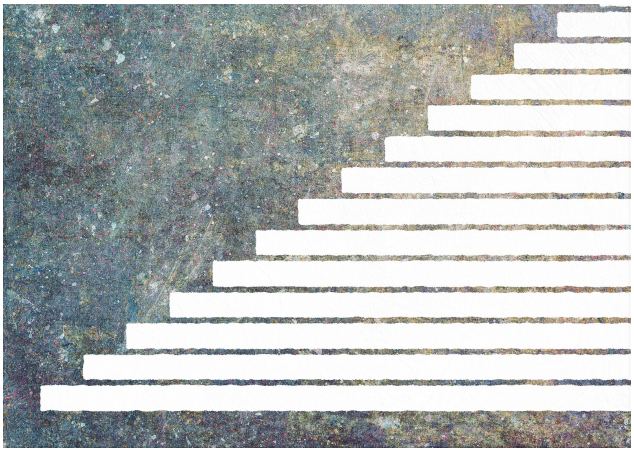
I once taught a group of students about instructional clarity using a set of sharpened pencils. If you have a set, grab them. If not, you can likely make it work just with imagination.

Take one pencil. (It must be sharpened for this to work.) Put your hand out, palm up. Now gently poke your palm. Notice how it felt. Now take 10-15 sharpened pencils, again, sharpened. Hold them together, again with the palm up. Now gently poke your palm. Notice how it felt.

One makes me flinch. 15 don't. This is a lesson about instructional clarity: in order to say more, we often need to say less. In order to ensure that our students learn more, we need to cover less content. One point, well-made, will stick. Fifteen points, even well made, will not.

Too often I am trying to jam many good points into student brains. It doesn't ever work. Learning happens when teaching is focused.

Let me add: my anecdotal experience around emotional health, rigor, and clarity: instructional clarity is more powerful at reducing anxiety than nearly any other strategy I know of. In particular, my students have communicated to me repeatedly that it is unclear expectations--not rigor--that are the cause of the bulk of their frustrations.



Practices:

Clarity Practice 1:
Busywork Audit
I asked students to rate my assignments on a 1-to-10 scale from "meaningless busywork" to "highly meaningful work." The results were eye-opening. Use your own students, or if not, hire a Student Consultant on Teaching (SCOT). They're pretty fantastic, and give great unbiased feedback.

Fewer Spinning Plates
I've switched from many assignments to few (but the same amount of work) and was surprised to find that students wer relieved: 10 weekly assignments are easier to keep track of than the same work broken into 40 assignments. Sometimes one large spinning plate is easier to keep in the air than many small ones.

Clarity Practice 2:
Art note: The title of the second piece of art is "Peace, Be Still." The art suggests to my mind the way that the Savior brings peace--which is full of implications if I intend to teach as He does.

Principle 3: Growth Mindset, and Progressive Instruction. "Line Upon Line"

"While I was a teacher, a principal taught me that I should always decorate my room in such a way that the classroom itself was teaching anytime students looked around. Brilliant advice, and I highly recommend it. Here's a reformulation: 'Decorate your office such that the walls preach sermons that words cannot.'"
--Letter to self, ¶ 4, Feb. 2021

My dad once bragged to me about finishing Isaiah again. (He reads it annually.) I was an obnoxious teenage type, so I retorted back, "Yeah, but did you understand any of it?"

His answer: "Oh, only a little. But I'm not afraid of it anymore."

Recently, a student of mine admitted that she was frightened of math. She asked for help. I talked about growth mindset. I talked, too, about the need to get prerequisite understanding down pat. I introduced her to Khan Academy, explained the way that Khan is modelled on mastery learning and growth mindset (though, admittedly, I did not explain as well as Sal Khan himself does) and began tutoring her.

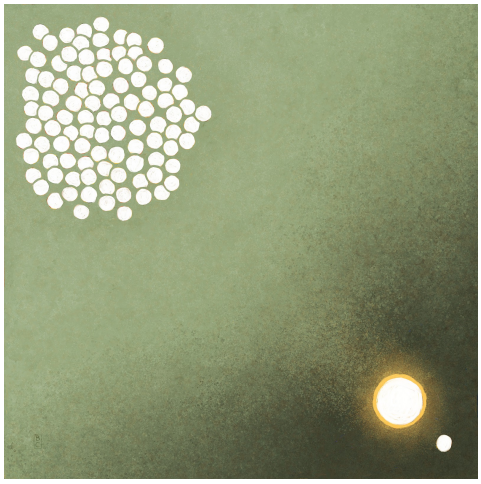
She returned a few days later and said, simply, "I'm not scared anymore. I don't have it all down, but I realize now that I can get it with practice. Mistakes are just things I haven't learned yet. I've never been so excited to get a problem wrong before, because I can learn from it."

Line-Upon-Line Practice 1:
Growth Celebrations
While teaching in middle school, I switched from having a pizza party for top test-scorers, and started inviting students who had grown the most. Two things happened: first, more students started trying, because the incentive was focused on growth, rather than mere achievement. Second, I spent more on pizza.

Line-Upon-Line Practice 2:
Delicious Mistakes, and Culture of Error
Once, in a middle school math class, I encouraged students to solve the problem correctly, and then to spot the most attractive wrong answer of the multiple-choice set. Students loved the chance to be "smarter-than-the-test," but they also embraced the idea of making good mistakes early.

Art note: The third painting's title is "Line Upon Line." The Doctrine and Covenants tell us that Jesus Himself did not receive of the fullness at first, but had to progress "grace for grace." I hope it suggests something about progression, the nature of healthy error, and of learning things "grace for grace"--what the world would call "growth mindset."

Principle 4: Individualized Learning. "One By One."
When students come to my office, I regularly receive impressions like "This one needs to know that they are loved" or "This one needs to see their own unique talents." Sometimes it's as simple as "This one just needs to laugh." I have yet to receive the impression "This one has enough light in their life. No need to say something uplifting."



It strikes me that our students do not have a surfeit of edification in their lives right now.

--Letter to self, ¶ 5, Feb. 2021

Recently, a friend reminded me that we don't teach classes of 40. We teach individuals. We may teach 40 individuals, but we teach one by one, as the Savior did.

The Savior invited the Nephites to come to Him "one by one." Perhaps more importantly, He laid hands upon the little Nephite children, and blessed them "one by one."

Mastery learning changes education from "all" to "each." It's about student pacing, yes, but it's about seeing them--their potential and their divine gifts--as God does.

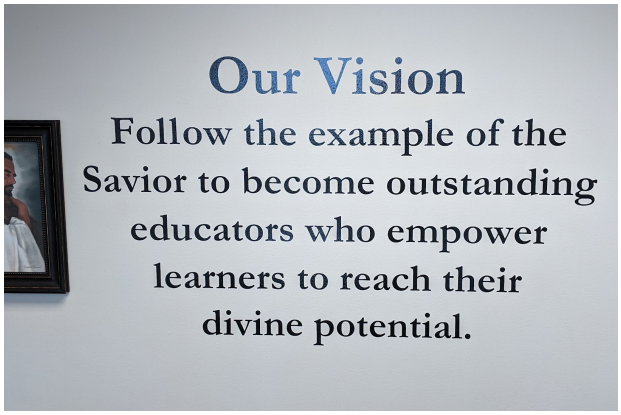
Personalization Practice 1:
Self-Paced Learning
When I was in fourth grade, my teacher let me skip all the homework for math if I could pass the chapter test with a B or better. I finished two full years of math in a single year (with a friend's help). I was no genius, but I sure felt like one. I could go slower if I needed more time on a challenging concept, but quicker when I already knew something. This is the heart of mastery learning.

Personalization Practice 2:
Learning and Identity
This practice is really a whole family of practices. I ask my students to reflect on why God would have them take this class at this particular time. What does it have to do with their life's mission? What does it have to do with unlocking their potential? What does their patriarchal blessing say about it? I deliberately aim to have them tie together their learning and their unique identity as children of God.

Art note: Painting four is entitled "Ninety and Nine II."

A Vision of Potential
"Those who need these messages--those who are straining hardest to hear an unsaid word--seem more willing to pay the price to understand. They search the figures in the paintings carefully, sensing some meaning behind them. As they search, they see barely-visible hand-writing (my wife's careful script) and the title of the piece on the matte under the painting. I love this part. I always try to watch their faces as understanding dawns."
--Letter to self, ¶ 6, Feb. 2021

Benjamin is a hero of mine. He was a connoisseur of brilliance--not merely brilliant himself, but interested in the stuff as a matter of research. He edited a book, called *Developing Talent in Young People*, in which he studied the unique capacities of unusually talented youth, and tried to describe how we could better foster the incredible potential of humankind. His system wasn't about rigor, clarity,



or pacing, it was about seeing the potential of each person.

You likely know him as Benjamin Bloom.

On the wall of the Hinckley building, you can find the vision statement of the elementary education department carefully inscribed, set tastefully next to a portrait of the Savior. It reads, “Our Vision: Follow the example of the Savior to become outstanding educators who empower learners to reach their divine potential.”

It was one--of many--quiet signals from God that this was the right place for me.

Mastery learning is the legacy of Benjamin Bloom. It works because it is based on true principles, and none truer than this: the truest aim of education is to unlock the brilliance within each learner.

And they are brilliant--if we’re willing to set the stage for them to shine.

All artwork from www.bencrowder.net/art.